Margarita Kattsyna

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SUMMARY

Software Engineer with hands-on experience designing and building scalable backend and full-stack systems using Spring Boot, React, Redux, TypeScript, Node.js, and modern web technologies. Proven ability to own features end-to-end, using my accurate and creative skills to produce high-quality code for various products or software services.

TECHNICAL SKILLS

- **Languages**: Java, JavaScript, TypeScript, C#
- **Frameworks/Libraries**: Spring Boot, React, Node.js, Express, Tailwind CSS, Vite, Redux, Zenject, XR Interaction Toolkit, OpenXR
- **Databases**: MySQL, PostgreSQL, MongoDB, SQLite, Relational databases
- **Tools & Platforms**: Git, GitHub, GitHub Actions, Docker, Sapling, Jira, CI/CD workflows, Unity
- **Core Concepts:** Data Structures & Algorithms, REST API Design, Backend Systems, Performance Optimization, Agile Development

PROFESSIONAL EXPERIENCE

VR/XR Software Engineering - Glendale Community College, Glendale, CA **Feb. 2026 - Present**

- Engineered interactive VR applications using Unity and industry-standard development tools that enhance student engagement, and deepen conceptual understanding.
- Collaborated in Agile teams to deliver iterative features, contributing to scalable system design.
- Designed and implemented testing workflows to improve system reliability and reduce runtime errors.

Software Engineering Tutor - Glendale Community College, Glendale, CA **Sept. 2025 - Feb. 2026**

- Mentored students in data structures, algorithms, and full-stack web development, reinforcing best practices in problem-solving and clean code.
- Guided students in debugging, complexity analysis, and system design fundamentals.

Software Engineering Intern - a startup TerryTech LLC, CA **Sept. 2025 – Dec 2025 17, 2025**

- Delivered production features for a cross-platform iOS application using Swift and Firebase, including CRUD functionality, onboarding flows, resulting in improved onboarding conversion and reduced user drop-off.
- Diagnosed and resolved UI and backend logic issues **by 50%**, improving UX, and system stability.
- Built complex front-end solutions to support high-traffic environments and unique user needs, with strong test coverage using modern abstractions and frameworks; leveraged experience with centralized version control.

RELEVANT PROJECTS

Banking Platform - Backend Java Application (Java, Spring Boot, Docker, Flyway)

- Built a REST back-end for accounts, money transfers and authentication with Spring Boot, structured as vertical slices with hexagonal architecture (domain, application ports, infrastructure adapters).
- Implemented concurrency-safe transfers with optimistic locking and idempotency keys, publishing domain events, and verified behaviour under concurrent requests with dedicated concurrency tests.
- Implemented JWT authentication with rotating refresh tokens and reuse detection, BCrypt password hashing and Spring Security.
- Wrote automated tests: domain and handler unit tests, end-to-end API tests against real PostgreSQL via Testcontainers, and ArchUnit rules enforcing module boundaries.

VR Card Desk: Interactive 3D Learning Cards · Unity 6, C# github.com/prostodiva/desk

- Built a VR card system in Unity/C# where players grab cards by hand, inspect them with the trigger (cards fly into view and follow the player's head), browse the deck with the thumbstick, and flip cards to read descriptions, set in a modelled classroom.

- Architected a layered ports-and-adapters design across 6 assembly definitions with compiler-enforced dependencies, so XR hardware stays isolated in a single adapter and game rules never reference input or rendering code.
- Wired the system with Zenject dependency injection, factories for spawning, and one installer as the composition root, replacing static singletons.

2D Game: Eu-rika: Unity, C# available at **eu-rika.com**

- Designed and developed a puzzle game in Unity (C#), implementing level progression, scoring, save system.
- Shipped a downloadable Windows / macOS build, available at eu-rika.com

eu-rika.com – Web Application (TypeScript, Node.js, Unity, C#)

- Designed and implemented a web interface using high-quality code to showcase the Unity interactive game; partnered cross-functionally across teams to solve abstract challenges and deliver scalable product solutions.
- Implemented modular components and optimized user interaction flows for performance and usability

nfl-vacation.pages.dev - Full-Stack Application

- Developed and maintained a full-stack(front-end and back-end) travel planning application using graph algorithms to optimize stadium routes and scheduling logic.
- Implemented interactive data visualization and responsive UI for improved usability.

EDUCATION

Associate in Science in Computer Science & Web-Dev Certificate — Glendale Community College, CA
Bachelor Of Science in Food Science — KSUTE, Krasnoyarsk, Russia
M.S. in Software Engineering - Application submitted — CalState Fullerton, CA